ECE 340: Semiconductor Devices

Lecture 3: Crystal Structure

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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Recap of last lecture

- **Lattice, Lattice points & Basis**
- **Unit Cell & Primitive Cell**
- **Primitive Vectors & Basis Vectors**

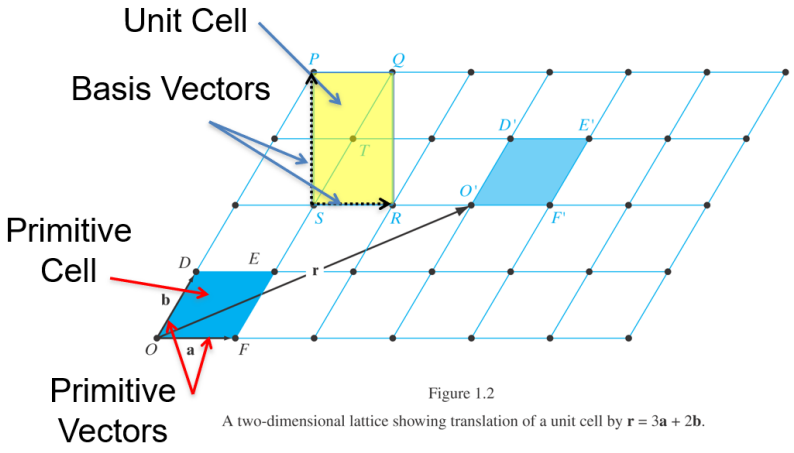


Figure 1.2

A two-dimensional lattice showing translation of a unit cell by $\mathbf{r} = 3\mathbf{a} + 2\mathbf{b}$.

Cubic crystal structure

- Simple cubic, BCC, FCC
- Diamond structure

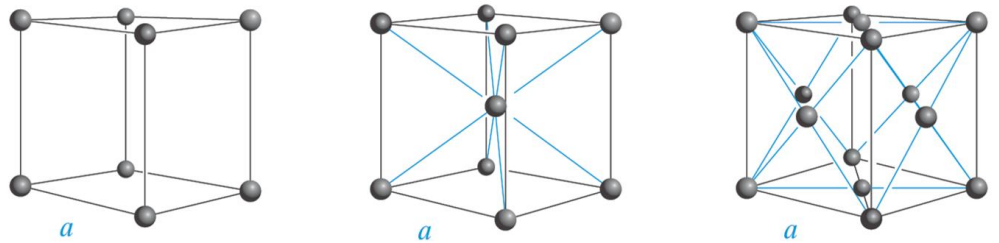
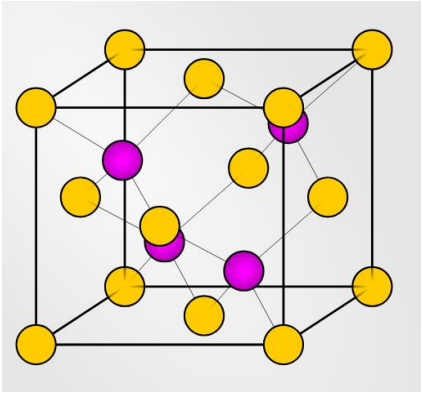


Figure 1.3

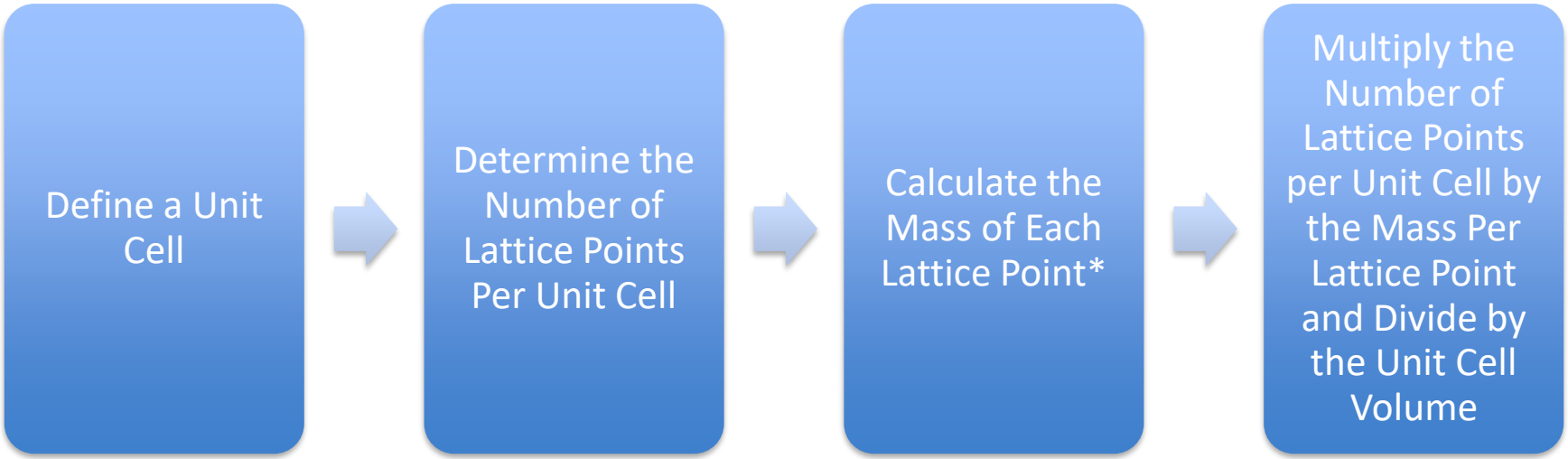
Unit cells for three types of cubic lattice structures.

Diamond structure:
Two superimposed FCC



Density Calculations

Cubic Crystal Density Calculations



*Each lattice point may consist of multiple atoms

FCC Lattice: How Many Lattice Points Per Unit Cell?

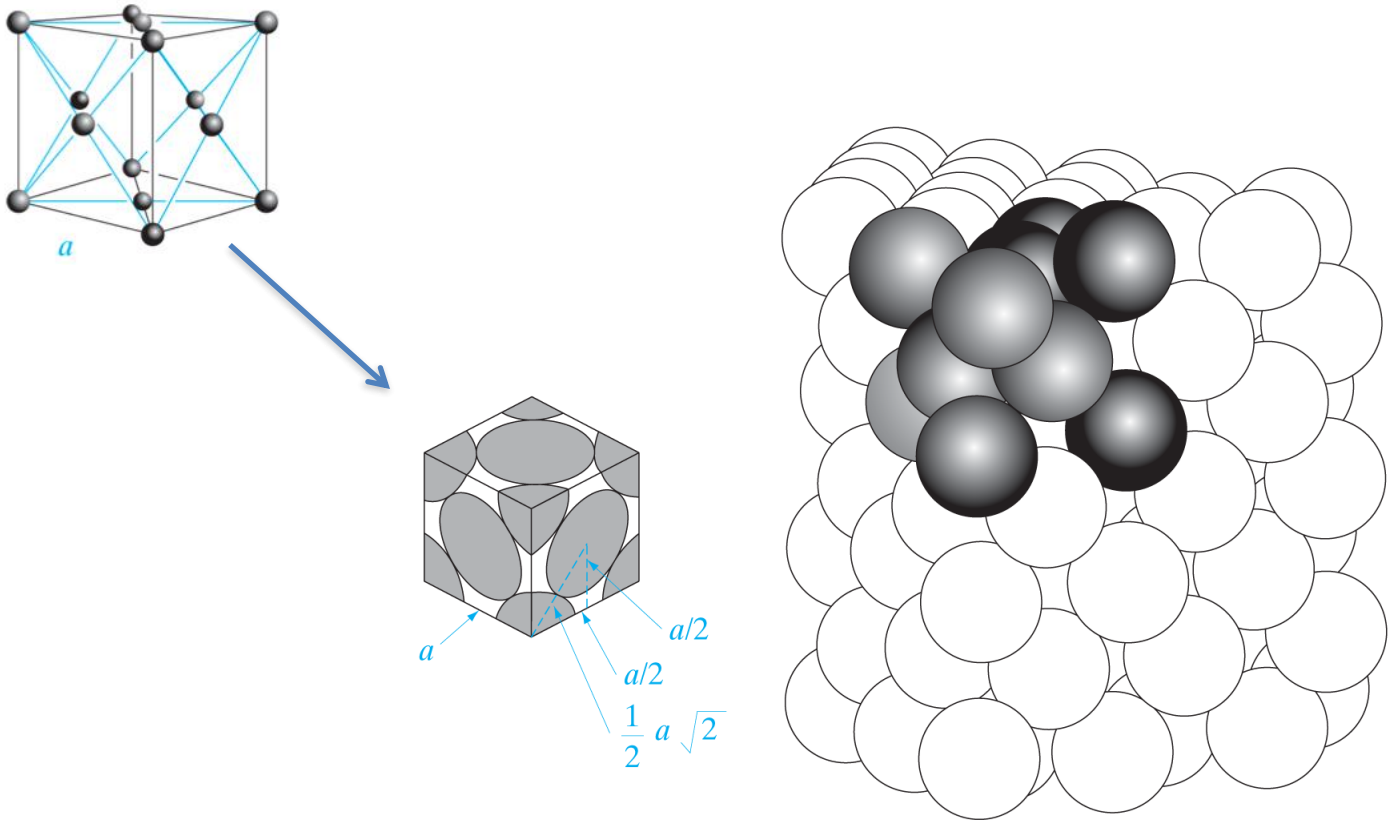


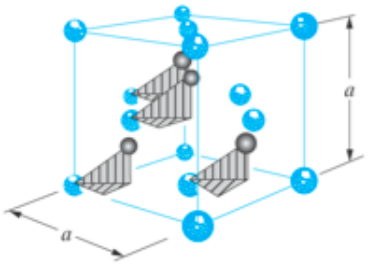
Figure 1.4

Packing of hard spheres in an fcc lattice.

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Example: Silicon

- The diamond lattice can be thought of as a face centered cubic (fcc) structure with an additional atom placed at 1/4 a +1/4 b + 1/4 c from each of the fcc atoms



- Volume Density of Silicon
  - Lattice Const = a_{Si} = 5.43Å = 5.43 x10^{-8} cm
  - Diamond Structure
    - 4 lattice points per unit cell, 2 atoms per lattice point, so 8 atoms/unit cell
    - 8 atoms/ $(a_{Si})^3$
    - = 5 x 10^{22} atoms/cm^3
  - Atomic weight for silicon= 28.1 (gram/mole)
  - Avogadro's Number = 6.02 x 10^{23} atoms/mole
  - Density of Silicon
    - = 5x10^{22} (atoms/cm^3) X 28.1 (grams/mole) X (1mole/6 x10^{23} atoms)
    - = 2.33 gram/cm^{-3}

$$\frac{atoms}{cm^3} \square \frac{grams}{mole} \square \frac{mole}{atoms} = \frac{grams}{cm^3}$$

Example: $Al_xGa_{1-x}As$

Define a Unit Cell

AlGaAs has a zinc blende lattice

The unit cell is FCC with 2 atoms per lattice point

The lattice spacing is determined using Vegard's Law (linear interpolation):

$Al_xGa_{1-x}As$ Lattice Spacing: $a_{AlGaAs} = xa_{AlAs} + (1-x)a_{GaAs}$

$Al_{0.25}Ga_{0.75}As$ Lattice Spacing: $a = (0.25)(5.6605) + (0.75)(5.6533) = 5.6551$ Angstroms

Determine the Number of Lattice Points Per Unit Cell: 4

Calculate the Mass of Each Lattice Point

Mass of As: $m_{AsAtom} = \frac{74.92g}{mole} \times \frac{mole}{6.02 \times 10^{23} atoms} = 1.24 \times 10^{-22} \frac{g}{atom}$

Mass of $Al_{0.25}Ga_{0.75}$: $\left(0.25 \times \frac{26.98}{6.02 \times 10^{23}}\right) + \left(0.75 \times \frac{69.72}{6.02 \times 10^{23}}\right) = 9.93 \times 10^{-23}$

Mass of Each Lattice Point: $m_{\text{LatticePoint}} = 1.24 \times 10^{-22} + 9.93 \times 10^{-23} = 2.23 \times 10^{-22} \frac{g}{\text{Lattice Point}}$

Calculate the Density

$$Density = \frac{4 \times \left(2.23 \times 10^{-22}\right) g}{\left(5.6551 \times 10^{-8} cm\right)^3} = 4.9 \frac{g}{cm^3}$$

Other Materials?

- Binary Zinc Blende?
- Ternary Zinc Blende?
- Quaternary Zinc Blende?
  - InAlGaP versus InGaAsP
- Wurtzite?

- Caution: Use Vegard's law to estimate properties such as lattice constant only if a specific reference is not available

Miller Indices

Determining Miller Indices

Crystallographic Notation
(hkl): Plane
[hkl]: Vector Normal to Plane
{hkl}: Family of Planes
<hkl>: Family of Vectors

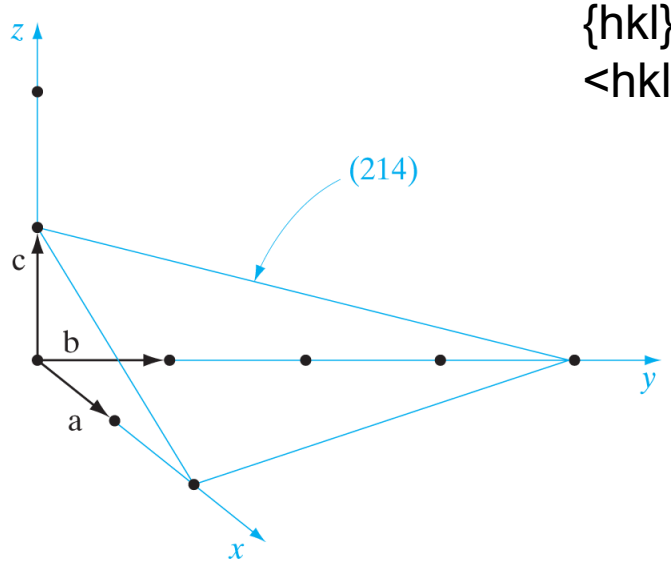


Figure 1.5
A (214) crystal plane.

Method
1. Determine intercept points
2. Take reciprocal
3. Multiply by least common multiple (smallest positive integer divisible by intercepts)

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Class Exercise

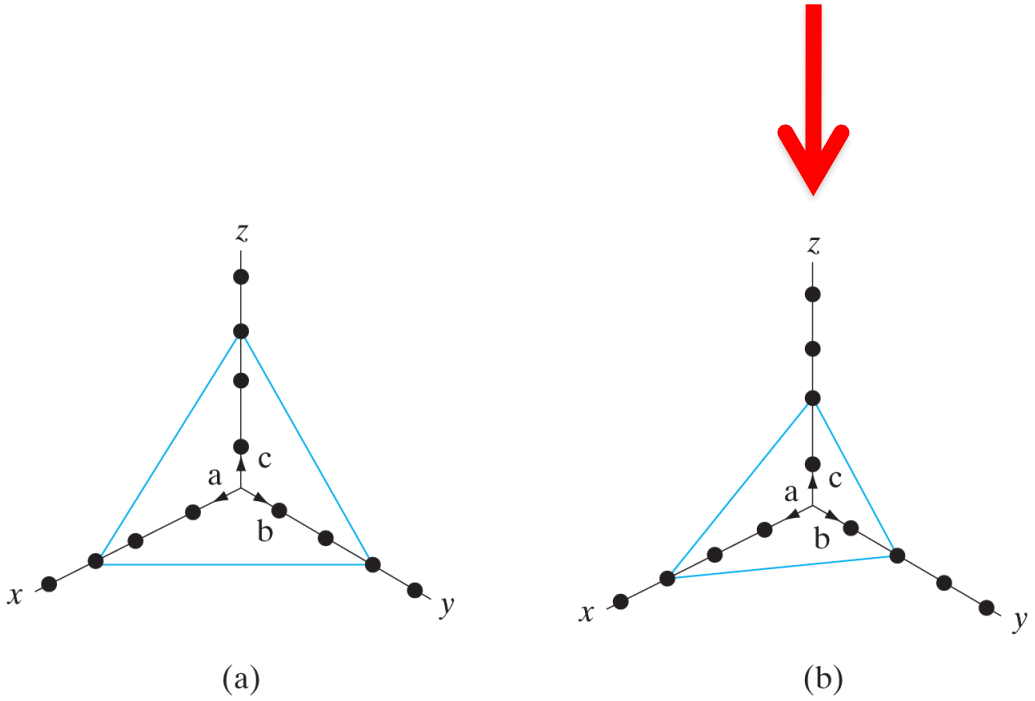


Figure 1.17.1 P1-3

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Direction Vectors

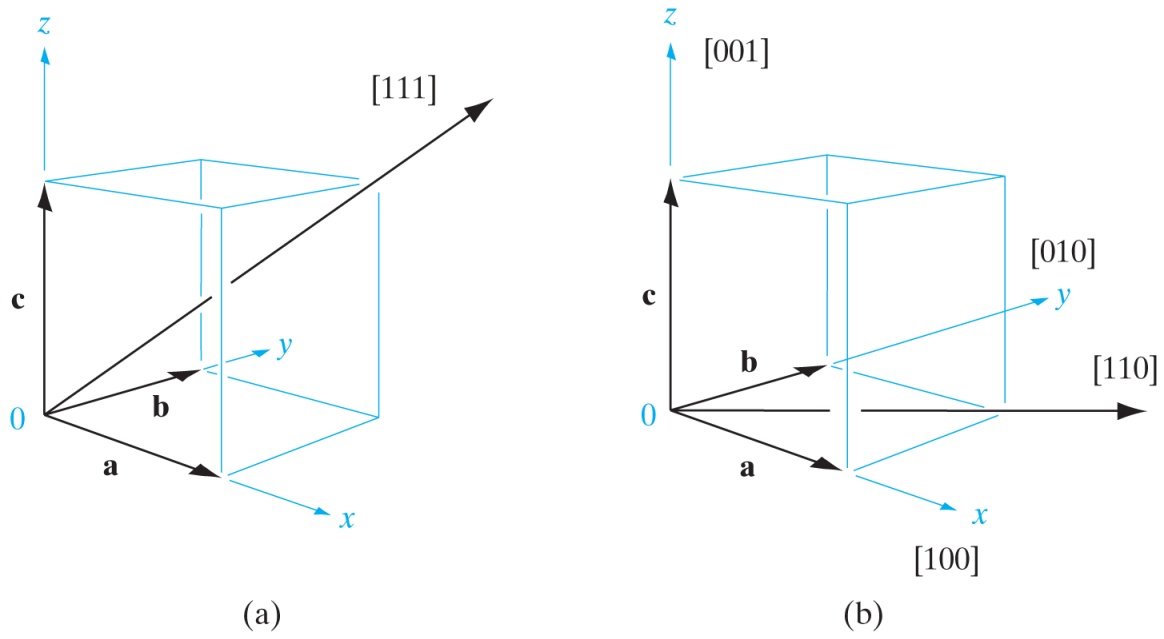


Figure 1.7

Crystal directions in the cubic lattice.

Key Property: Normal to equivalent plane for cubic lattices

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Plane Separation & Direction Angles

Distance Between Adjacent (h k l) Planes (lattice constant "a"):

$$d = \frac{a}{\left(h^2 + k^2 + l^2\right)^{1/2}}$$

Angle Between Directions $[h_1\ k_1\ l_1]$ and $[h_2\ k_2\ l_2]$:

$$\cos(\theta) = \frac{\left(h_1 h_2 + k_1 k_2 + l_1 l_2\right)}{\left(h_1^2 + k_1^2 + l_1^2\right)^{1/2}\left(h_2^2 + k_2^2 + l_2^2\right)^{1/2}}$$

Symmetries and Families of Planes/Directions

Symmetry Operations

- If an operation on the lattice (rotation, translation, inversion, etc.) produces an identical lattice, it is a symmetry operation
- Planes and directions are in the same family if they are related by a symmetry operation
- Simple cubic structures have high symmetry and many related families of planes
- Zinc blende (III-V) structures have lower symmetry, so planes that are in the same family for a cubic structure may not be in the same family in a zinc blende structure

Examples of Families of Planes

- The planes (100), (010), and (001) are all related by a symmetry operation for a simple cubic lattice, and are thus part of the {100} family
- For other crystals structures, (100), (010), and (001) may not be part of the {100} family

Crystallographic Notation
(hkl): Plane
[hkl]: Vector Normal to Plane
{hkl}: Family of Planes
<hkl>: Family of Vectors

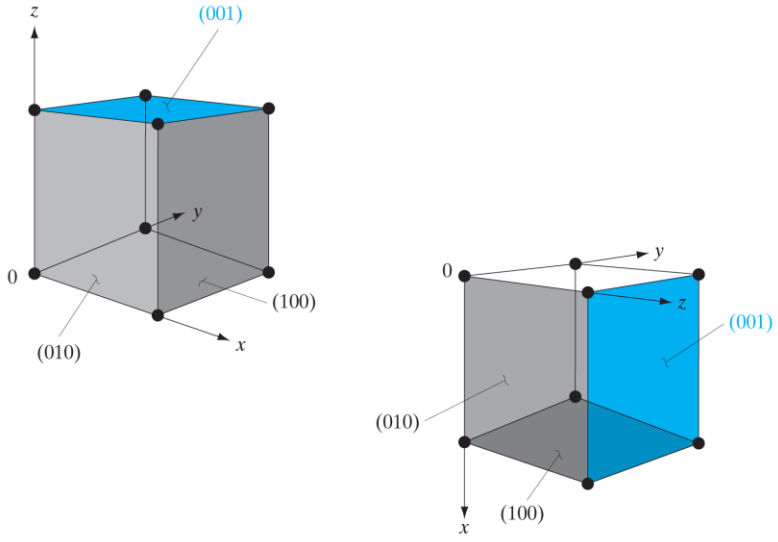


Figure 1.6
Equivalence of the cube faces ({100} planes) by rotation of the unit cell within the cubic lattice.

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Crystal Growth & Strained Layers

Silicon Purification

Formation of Metalurgical-Grade Silicon (MGS):

$SiO_2 + 2C \rightarrow Si + 2CO$ (~1800° C)

Purification of MGS (Removal of Al, Fe, heavy metals, etc.):

Step 1: Formation of Trichlorosilane (liquid, $T_{Boil} = 32°$ C)

$Si + 3HCl \rightarrow SiHCl_3 + H_2$

Step 2: Fractional Distillation

Step 3: Conversion of Trichlorosilane to Electronic-Grade Silicon

$2SiHCl_3 + 2H_2 \rightarrow 2Si + 6HCl$

Bulk Growth

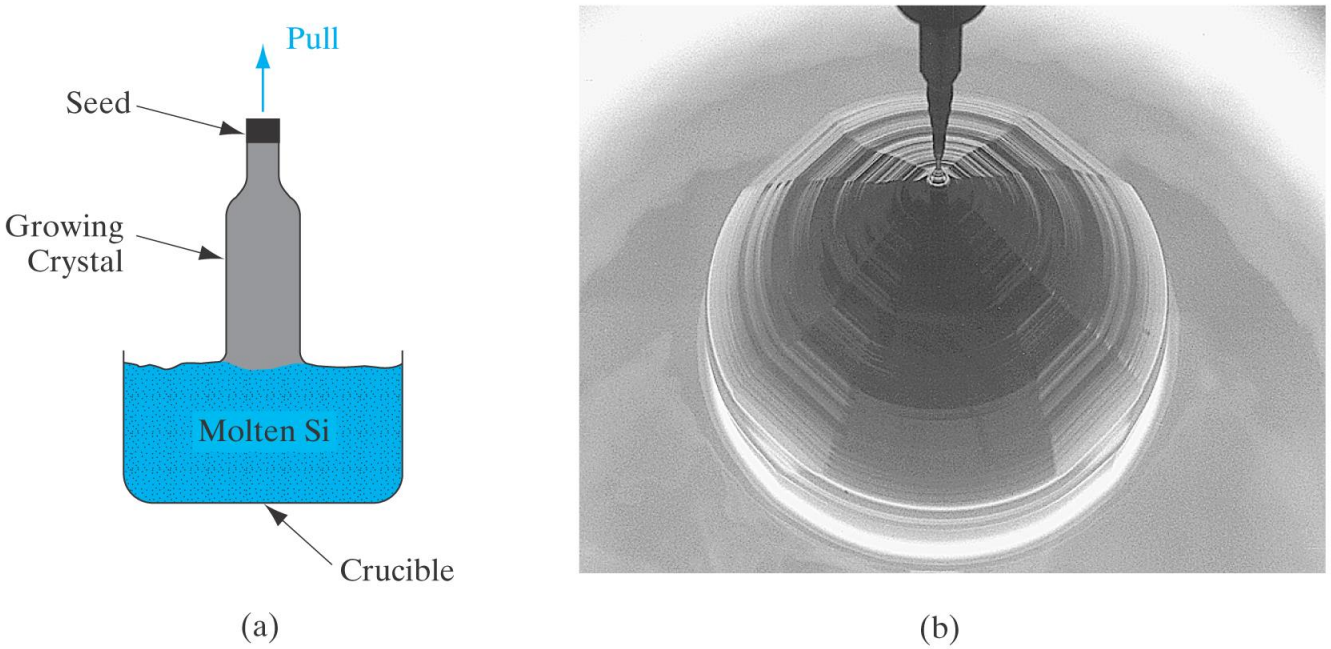


Figure 1.10

Pulling of a Si crystal from the melt (Czochralski method): (a) schematic diagram of the crystal growth process; (b) an 8-in. diameter, ⟨100⟩ oriented Si crystal being pulled from the melt.
(Photograph courtesy of MEMC Electronics Intl.)

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Silicon Ingot

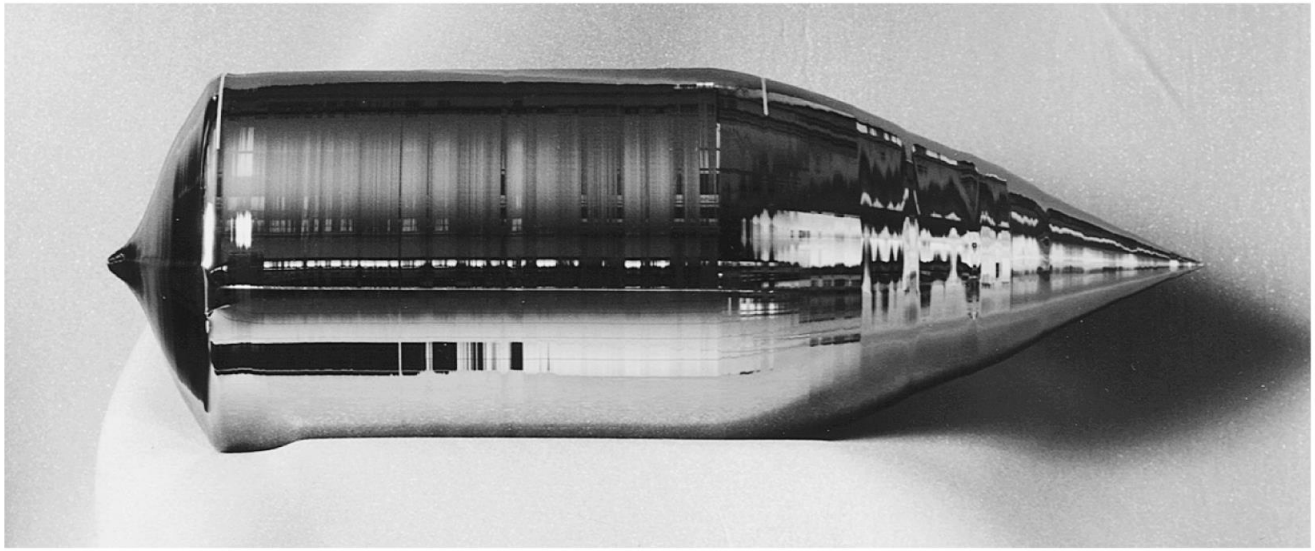


Figure 1.11

Silicon crystal grown by the Czochralski method. This large single-crystal ingot provides 300 mm (12-in.) diameter wafers when sliced using a saw. The ingot is about 1.0 m long (including the tapered regions), and weighs about 140 kg. (Photograph courtesy of MEMC Electronics Intl.)

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Epitaxial Growth

Lattice Matched Systems

Heteroepitaxy: Grown materials are lattice matched but have different properties than the substrate

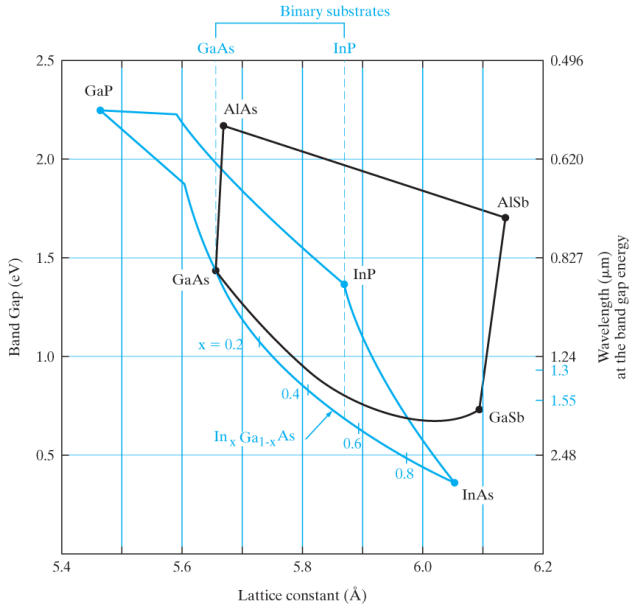


Figure 1.13

Relationship between band gap and lattice constant for alloys in the InGaAsP and AlGaAsSb systems. The dashed vertical lines show the lattice constants for the commercially available binary substrates GaAs and InP. For the marked example of $In_xGa_{1-x}As$, the ternary composition $x = 0.53$ can be grown lattice-matched on InP, since the lattice constants are the same. For quaternary alloys, the compositions on both the III and V sublattices can be varied to grow lattice-matched epitaxial layers along the dashed vertical lines between curves. For example, $In_xGa_{1-x}As_yP_{1-y}$ can be grown on InP substrates, with resulting band gaps ranging from 0.75 eV to 1.35 eV. In using this figure, assume the lattice constant a of a ternary alloy varies linearly with the composition x.

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Strained Layers

- Pseudomorphic Growth: Epitaxial growth of a non lattice-matched material

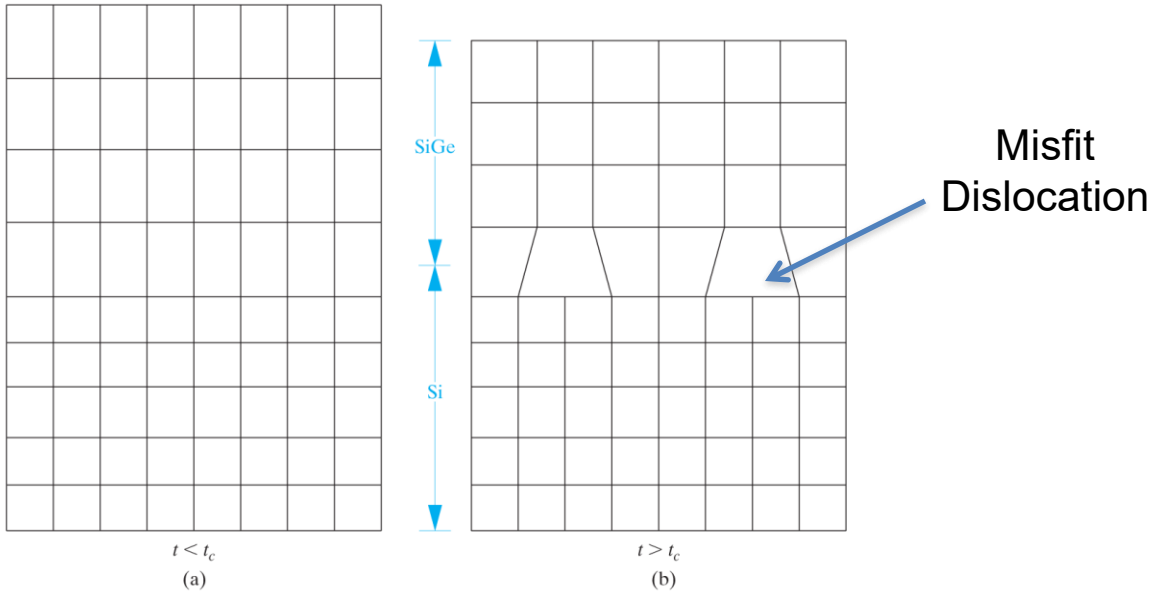


Figure 1.14

Heteroepitaxy and misfit dislocations. For example, in heteroepitaxy of a SiGe layer on Si, the lattice mismatch between SiGe and Si leads to compressive strain in the SiGe layer. The amount of strain depends on the mole fraction of Ge. (a) For layer thicknesses less than the critical layer thickness, t_c, pseudomorphic growth occurs. (b) However, above t_c, misfit dislocations form at the interface which may reduce the usefulness of the layers in device applications.

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Other Concepts

- **Critical Thickness:** Growth thickness above which misfit dislocations are formed
- **Strained Layer Superlattice:** A structure of alternating layers having tensile and compressive strain such that net strain balance occurs and misfit dislocations are not formed

Epitaxial Growth Methods

- Liquid Phase Epitaxy
- Vapor Phase Epitaxy
  - (Tri)Chloride & Hydride VPE
- Molecular Beam Epitaxy
- Chemical Vapor Deposition
- Metalorganic Chemical Vapor Deposition